



Geographic distribution of two *Montivipera* taxa using ecological niche modeling

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Ecological niche modeling predicts the potential distribution areas of species and contributes to measuring the niche differentiation among taxa. *Montivipera albizona* and *M. bulgardaghica* are two mountain viper taxa distributed in Anatolia. In the present study, the habitat suitability of *M. albizona* and *M. bulgardaghica* was assessed, and ecological niche ranges between these taxa were compared. According to species distribution models, while the distribution area of *M. albizona* covers a relatively large region in central Anatolia, *M. bulgardaghica* has a relatively narrow distribution over the eastern Taurus mountains. The species distributional results showed that the annual temperature range and mean temperature of the coldest quarter were the highest contributing variables, with 62.5% for *M. albizona* and 67.9% for *M. bulgardaghica*. *Montivipera albizona* prefers 13 different habitat types, while *M. bulgardaghica* prefers 6 different habitat types in their distribution areas. Niche differentiation analysis revealed that there was no overlap between the two taxa in terms of their ecological niches.

Keywords. Anatolia; habitat selection; *Montivipera*; niche differentiation; reptiles; Viperidae

1. Introduction

Reptiles are a relatively large group but have been less studied ecologically than some terrestrial vertebrate groups (i.e. birds and mammals) (Rodrigues *et al.* 2005; McCain 2010; Guedes *et al.* 2017). Since reptiles are generally characterized by narrow niche requirements and their locomotion ability is more restricted (Rodrigues *et al.* 2005; McCain 2010; Guedes *et al.* 2017), knowing their potential spatial distribution is very important and crucial. This distribution enables them to be used as suitable biological models for assessing speciation processes (Doan 2003; Agarwal *et al.* 2014). This provides information about the distribution of reptile groups with closely related and sympatric populations and gives us an idea of whether there is a conserved niche among these populations (Kafash *et al.* 2020).

Within the class Reptilia, snake species are susceptible creatures. They are disproportionately threatened

with extinction, mainly because they are either venomous or thought to be venomous (Maritz *et al.* 2016; Saha *et al.* 2018; Mebert *et al.* 2020). This has made it essential to evaluate the conservation status of these creatures, determine their distribution and their ecological niches, and understand their biology better. Mountain vipers (genus *Montivipera*) are unique creatures that generally inhabit places above 1000 m altitude, and as a result, they have adapted to mountainous conditions (Mebert *et al.* 2020). Since these creatures live at high altitudes, their annual activities are an essential life strategy for them. Additionally, their reproductive activities must be completed in less time than at low altitudes. This phenomenon reveals that mountain vipers need an exceptional niche compared with their relatives living at low altitudes.

The genus *Montivipera* is a complex species represented by five different recognized species (*M. albizona* Nilson *et al.* 1990, *M. bulgardaghica* Nilson and

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Andrén 1985, *M. raddei* Boettger 1890, *M. xanthina* Gray 1849, and *M. wagneri* Nilson and Andrén 1984) in Türkiye (Kurnaz 2020). The taxonomy and phylogeny of viper species belonging to this group have been studied for about two decades (Stümpel and Joger 2009; Stümpel *et al.* 2016). *Montivipera albizona* is a species that differs morphologically from the *M. bulgardaghica* taxon by the excess of lateral head scales and a different ridge pattern (Nilson and Andren 1985; Nilson *et al.* 1990). However, according to recent phylogenetic studies, *M. albizona* is nested in *M. bulgardaghica*, i.e., *albizona* is a subspecies of *bulgardaghica* (Stümpel and Joger 2009; Stümpel *et al.* 2016; Freitas *et al.* 2020). Although the distribution areas of these taxa are relatively distinct, there might be a contact zone between these taxa (Mebert *et al.* 2020). Although the contact zone is not at a suitable altitude (mostly lower than 1000 m a.s.l.) for both taxa to live, it is seen as the only region where gene flow can occur (Mebert *et al.* 2020).

As can be understood from the literature, no study provides information about the potential distribution, niche divisions, and habitat diversity of *M. albizona* and *M. bulgardaghica*. Hence, the present study is aimed at

1. determining the potential distributions of the two taxa under current conditions, using both bioclimatic variables and locality records
2. determining whether the niches acquired by the two taxa in their distribution areas are different from each other under the current conditions
3. examining whether these two species differ from each other in terms of the EUNIS (pan-European) habitat classification.

2. Materials and methods

2.1 Study area and occurrence data

Montivipera albizona and *M. bulgardaghica* taxa are generally distributed in the Anatolian Diagonal (Baran *et al.* 2021). The Anatolian Diagonal is an important geographical feature that starts from the eastern Black Sea mountains (near Bayburt-Gümüşhane) in the north and extends to the Amanos mountains and the eastern Taurus mountains in the Hatay province, forming a geographic isolation region between the species (figure 1a). The Anatolian Diagonal includes the Central Anatolian Region, the western part being below 1500 m and the eastern part above 1500 m elevation

(Gür 2017). Of the two species included in this study, the *albizona* taxon is generally adapted to live in the eastern parts of central Anatolia, and the *bulgardaghica* taxon is adapted to live in the higher regions of the eastern Taurus mountains. These areas have suitable habitats and features that reveal ecological niche requirements for the two species to survive. Therefore, the distribution area used in this study was limited to 33–40° east longitudes and 36–40° north latitudes, considering the areas where the *M. albizona* and *M. bulgardaghica* taxa live in the Anatolian Peninsula (figure 1b). A total of 69 occurrence records (47 for *M. albizona* and 22 for *M. bulgardaghica*) were gathered from the literature (Nilson and Andrén 1985; Nilson *et al.* 1990; Teynie 1991; Mulder 1995; Göçmen *et al.* 2009, 2014; Coşkun *et al.* 2012; Stümpel *et al.* 2016, 2019; Mebert *et al.* 2020) to perform Ecological Niche Modeling (ENM). All occurrence records are provided in supplementary table 1. Data for these species were error-checked and improved to meet appropriate standards for ENM in two steps. First, geo-referenced data were checked for error and data consistency for geographic coordinates (Chapman 2005). Second, to avoid spatial sampling biases and misinterpretation of the habitat suitability analysis and niche overlap tests, the occurrence records for each species were spatially rarefied in 20 km by SDM Toolbox 2.0 (Brown 2014).

2.2 Preparing variables

In order to establish the distribution model of the species, 19 bioclimatic variables (with 30 arc seconds resolution in the 2.1 version) were downloaded from www.worldclim.org (Fick and Hijmans 2017) (supplementary table 2). Additionally, habitat classification variables were downloaded from <https://land.copernicus.eu/> (European Nature Information System, hereafter EUNIS). Each bioclimatic and habitat variable covering the whole world (30 arc seconds (~1 km) resolution) was matched to the terrestrial regions of the study area where the two taxa were distributed using the Arc Toolbox embedded in ArcGIS version 10.7. All scores of the variables were exported, and Pearson correlation between the variables was calculated using the SPSS 21 software (IBM Inc.), and highly correlated parameter pairs ($r > |0.745|$) were excluded from the analysis to eliminate adverse consequences (supplementary table 3).

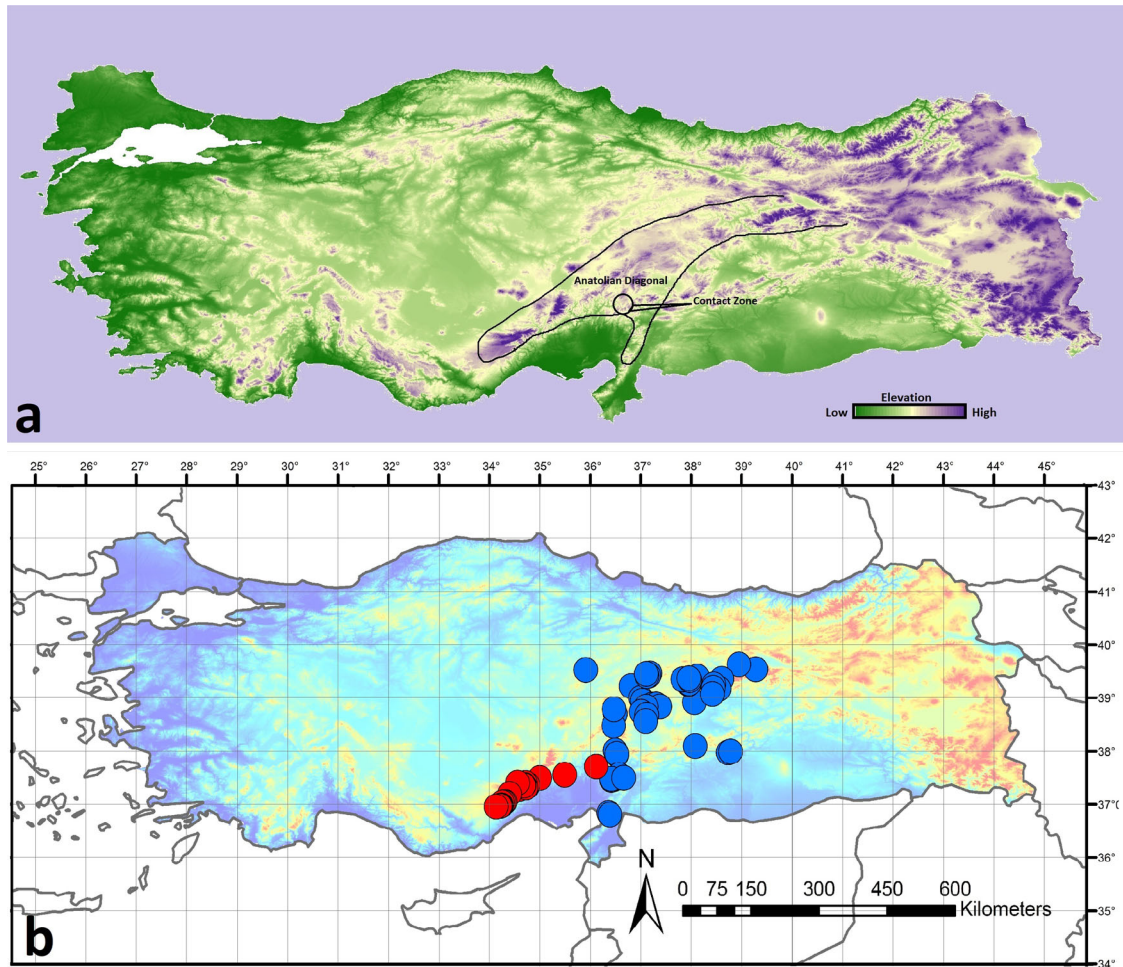


Figure 1. The maps show the Anatolian Diagonal and contact zone of two taxa (a), and species occurrence records for *M. albizona* (blue circles) and *M. bulgardaghica* (red circles) in the Anatolian Peninsula (b).

2.3 Distribution analyses and model selection

The MaxEnt software (v3.4.1) was used to calculate the potential distribution of the two mountain viper species (Phillips *et al.* 2017). To ensure that potential distributions of these two taxa are close to normal, 80% of the entity data were used for model training and the remaining 20% of entity data were used as test data (Cobos *et al.* 2019). MaxEnt, among the most effective algorithms for ENM, was run with the following settings: create response curves (showing how the prediction depends on a particular bioclimatic variable), cloglog output format, 20% random test percentage. A total of 7000 background points for each species was randomly sampled across the study area.

An R package that uses the MaxEnt program, *kuenm*, was used to generate the model calibration, calculate the distribution analysis, and evaluate the resulting data. The MaxEnt software uses two main modifiable parameters: (1) editing multiplier and (2) feature

classes (Cobos *et al.* 2019). The regulation multipliers consist of 17 values (0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 2, 3, 4, 5, 6, 8, 10). The feature classes consist of 5 classes: linear (l), quadratic (q), product (p), threshold (t), and hinge (h). The 17 regulation multipliers were tested for each species. Then, the 5 feature classes in the MaxEnt model were edited, and 31 different combinations of 17 different feature classes (l, q, p, t, h, lq, lp, lt, lh, qp, qt, qh, pt, ph, th, lqp, lqt, lqh, lpt, lph, lth, qpt, qph, qth, pth, lqpt, lqph, lqth, lpth, qpth, lqpth). Thus, a total of 527 different candidate models were produced for each species. Models with the lowest AICc and delta_AICc values were selected to determine the best fit among the calculated models (Akpoti *et al.* 2020; Arenas-Castro *et al.* 2020). In addition to AICc, AUC scores were also used to assess the accuracy of the distribution models. According to Manel *et al.* (2001), model scores were assessed as follows: AUC=0.5 reflects a performance equivalent to random, AUC>0.7 reflects a useful performance, AUC>0.8

reflects a good performance, and $AUC \geq 0.9$ reflects an excellent performance.

2.4 Niche differentiation analyses

Niche differentiation analyses were performed using the ENMTools 1.3 software (Warren *et al.* 2010). This software was used in identity and niche overlap tests, commonly used for evaluating the differences between the potential habitat suitability scores of both mountain vipers. The niche identity test allows the user to test whether the habitat suitability scores generated by ENM models from two species exhibit statistically significant ecological differences. The overlap analysis measures the similarity between habitat suitability predictions between one or more pairs of populations/species using the methods introduced by Warren *et al.* (2008). Two indices were measured to assess the niche identity of the species. These are Hellinger's I (Schoener and Gorman 1968) and Schoener's D (Warren *et al.* 2008). The indices' scores varied between 0 (completely disjointed) and 1 (identical). In order to examine the significance of niche differentiation, 100 pseudo-replicates were generated with a pool of occurrence records of the studied species. Finally, the indices' scores of the observed niche overlap (D_{H0} and I_{H0}) and identity test (D_{H1} and I_{H1}) were compared. The confidence level for D_{H1} and I_{H1} was 95% for this comparison (Warren *et al.* 2010).

2.5 EUNIS habitat types of the species

In addition to ENM analyzes, EUNIS habitat classes were used to calculate the similarities or differences in the habitat diversity used by these species in their distribution areas. The EUNIS habitat classification is an extensive organizational structure covering the European landmass terrestrial and marine habitat types and surrounding seas. Habitat types are characterized using a system of parameters, i.e., type of substrate, dominant life form, humidity, anthropogenic usage, and impact (Moss 2008). To know the EUNIS habitats,

the ARC-GIS compatible shapefile was applied to the study area, and the habitat types of the locality points were determined (www.eea.europa.eu).

3. Results

ENM enables researchers to analyze species distribution with even a small number of occurrence records, and, therefore, the number of occurrence records collected for both species was sufficient to perform dispersion analyses (Hosseinizadeh *et al.* 2018; Kafash *et al.* 2019; Heidari 2021). The distribution models for *M. albizona* and *M. bulgardaghica* were constructed using a set of 7 bioclimatic variables and 1 habitat variable because these were under the critical threshold value ($r < 0.75$): Mean Diurnal Range (Bio 2), Isothermality (Bio 3), Maximum Temperature of Warmest Month (Bio 5), Temperature Annual Range (Bio 7), Mean Temperature of Coldest Quarter (Bio 11), Annual Precipitation (Bio 12), Precipitation of Driest Quarter (Bio 17), and EUNIS. Based on the distributional analyses, it was found that Bio 11 (36.5 %) for *M. albizona* and Bio 7 (26%) for *M. bulgardaghica* were variables with the most contribution in the current climatic conditions. The contributions of the variables are given in table 1.

For *M. albizona* and *M. bulgardaghica*, ENM performed much better than a random prediction. The higher MaxEnt model performances (AUC and AICc values of 0.899 ± 0.037 and 1099.489 for *M. albizona*; 0.962 ± 0.035 and 467.583 for *M. bulgardaghica*) indicate that the potential distributions of both mountain viper species under current bioclimatic conditions are highly correlated with their occurrence records (table 2). The predicted model showed high AUC performance for the potential distribution of both species. The fact that the AUC data were very close to 1 showed that the distribution area revealed by the locality data obtained from the distribution areas of the two taxa was compatible with the estimated distribution areas obtained as a result of the analysis. As is known, while *M. albizona* inhabits the Anatolian Diagonal area, *M. bulgardaghica* spreads in a limited

Table 1. Percentage contribution of all examined layers in species distribution modeling of *Montivipera* species

Species	Bio 2	Bio 3	Bio 5	Bio 7	Bio 11	Bio 12	Bio 17	EUNIS
<i>M. albizona</i>	3.2	4.9	2.6	26	36.5	8.7	11.1	6.8
<i>M. bulghardaghica</i>	0	0	3.8	45.5	20	22.4	1	7.2

Table 2. Table for selecting the best model for species distribution maps of *Montivipera* species via the *kuenm* package

Species	All candidate models	Statistically significant models	Mean AUC ratio	AICc	W_AICc	delta_AICc	AUC
<i>M. albizona</i>	527	526	1.701	1099.489	1	0.052	0.872 ± 0.059
<i>M. bulghardaghica</i>	527	525	1.892	467.583	1	0	0.968 ± 0.030

area in the high places of the eastern Taurus mountains. The output of ENM reported a continuous prediction of bioclimatic suitability of specific geographical locations for both mountain vipers.

The distribution analysis results showed that the potential habitats calculated for both taxa were consistent with the occurrence records under the current bioclimatic conditions. (dark areas in figure 2). According to the analysis outputs, while the distribution area of *M. albizona* covers a relatively large region in central Anatolia, *M. bulghardaghica* has a relatively narrow distribution over the Taurus mountains. This might result from a more local distribution pattern of *M. bulghardaghica*.

Ecological niche divergence results showed no niche overlap between these two species (Hellinger's $I=0.604$ and Schoener's $D=0.278$ for *albizona/bulghardaghica*) for the climatic conditions. The identity test showed that the null hypothesis was rejected (t -test, $df=99$, $P<0.05$), and as a result, there was no niche overlap between the two species under current climatic conditions. The model demonstrated that the estimated niche model for *albizona/bulghardaghica* ($D_{H0}=0.726\pm0.037$ vs. $D_{H1}=0.278$ and $I_{H0}=0.919\pm0.015$ vs. $I_{H1}=0.604$) in the climatic conditions (figure 3) were completely differentiated. According to the identity test, there was no overlap between the two species. Besides, the model revealed that estimated niche models for the two species were completely separated and significantly distinct for the climatic conditions.

On the other hand, the EUNIS habitat classification analysis demonstrated that these two species used 13 different EUNIS habitats in their distribution area. While 6 of them have been used by both species (sparsely vegetated areas, transitional woodland–shrub, bare rocks, coniferous forest, natural grasslands, land principally occupied by agriculture, with significant areas of natural vegetation), the remaining 7 habitat types were solely used by *M. albizona* (mixed forest, complex cultivation patterns, permanently irrigated land, non-irrigated arable land, discontinuous urban fabric, broad-leaved forest, sclerophyllous vegetation) (supplementary figures 1–2).

4. Discussion

Two important data types are frequently used in studies aiming to determine the estimated distribution of species (Phillips *et al.* 2006): one is bioclimatic data, and the other is species' locality data (Phillips *et al.* 2006). Additionally, the number of occurrences is crucial to determine the possible distribution of the species (van Proosdij *et al.* 2016). In a study in which the number of minimum occurrences for species distribution models was calculated, it was reported that 25 occurrences would be sufficient, depending on whether the species has a widespread or narrowly distributed pattern (van Proosdij *et al.* 2016). Moreover, there are many ENM studies where the number of occurrences is close to or less than the number used in this study (Hosseinzadeh *et al.* 2018; Kafash *et al.* 2019; Taucare-Ríos and Piel 2021; Kurnaz 2022). Therefore, the number of occurrences used in this study was sufficient to perform the analysis for this study.

The traditional model of niche conservation of species is explained by the conservation of ancestral forms and niche evolution (Wiens and Graham 2005; Wiens *et al.* 2010). Niche conservation is more stable, especially in populations of allopatric species, as there is a limited flow of genes and association with other species (Wiens and Graham 2005; Wiens *et al.* 2010). Species in the genus *Montivipera* are generally allopatric, and almost all have niche preservation after allopatric speciation (Ahmadi *et al.* 2021). Since it is believed that there is a contact zone between the geographical distribution areas of two taxa that are the subject of this study, both of them are considered parapatric taxa (Meber *et al.* 2020). The fact that this contact zone is not at a suitable altitude for both *albizona* and *bulghardaghica* species to inhabit shows that this region is the isolation point between two taxa and that these two populations are allopatric rather than parapatric (Ahmadi *et al.* 2021). The ecological niche findings from this study also show that this geographical difference between these taxa might cause ecological niche differences. Furthermore, a study including the phylogenetic niche difference mentioned that these

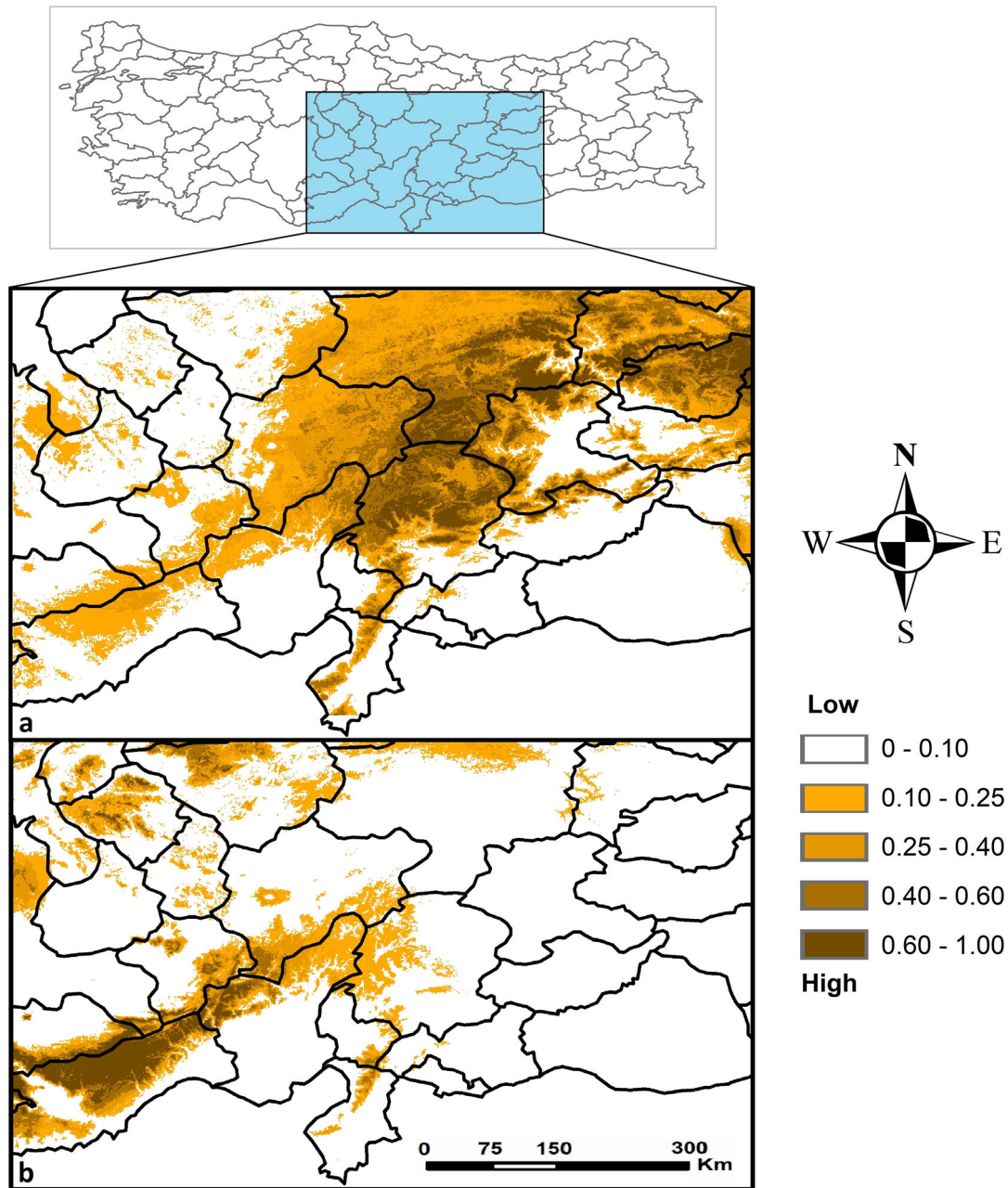


Figure 2. Habitat suitability predictions of (a) *M. albizona* and (b) *M. bulgardaghica* in Anatolia under the present climatic conditions. The blue area represents the area where the distribution analysis of the two taxa was performed.

two taxa showed allopatric speciation and that the Anatolian Diagonal caused vicariance between these taxa (Ahmadi *et al.* 2021).

ENM provides important opportunities to aid the niche differentiation dynamics among closely related groups and to speculate on whether they may be species or subspecies (Nakazato *et al.* 2010; Hosseinian Yousefkhani *et al.* 2016; Kurnaz and Hosseinian Yousefkhani 2020, 2021; Heidari 2021; Kurnaz and Şahin 2021; Şahin *et al.* 2022). It is a fact

that each species has different ecological requirements, and needs different ecological requirements in its habitats to be different from other species (Shapiro *et al.* 2016; Kurnaz and Eroğlu 2020; Şahin *et al.* 2021). This is one of the crucial factors in the differentiation of the close taxa. *Montivipera albizona* and *M. bulgardaghica* are valid taxa according to the study of Stümpel *et al.* (2016). Besides, it was reported that *M. albizona* was nested in the *M. bulgardaghica* clade and should be reduced to a

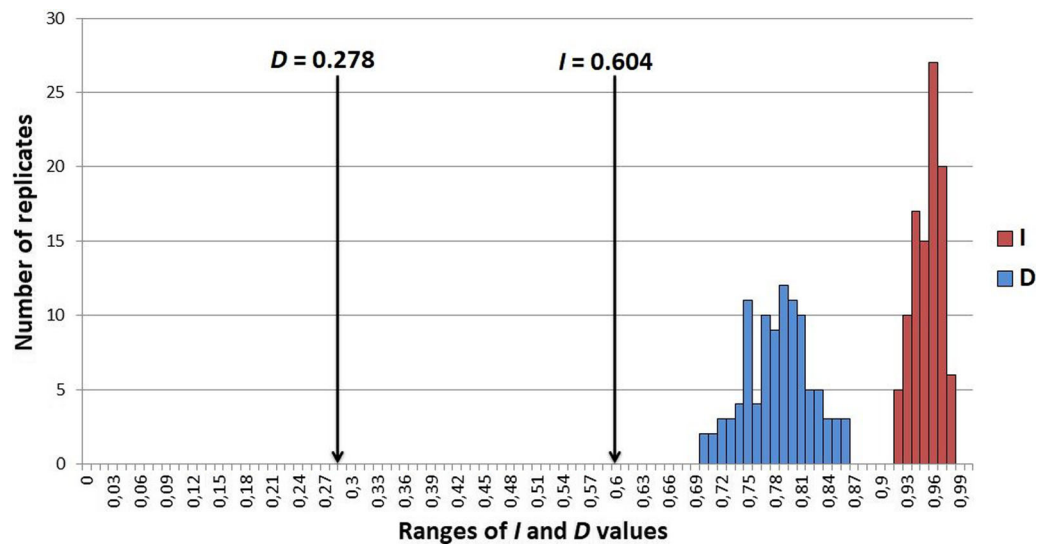


Figure 3. The identity test (D and I) results under the present climatic conditions. The bars with different colors are calculated as the significance threshold of the replicates with identity test mode, and arrows refer to actual niche overlaps.

subspecies level based on the results of recent phylogenetic studies (Stümpel and Joger 2009; Stümpel *et al.* 2016; Freitas *et al.* 2020). Ecological niche differentiation, one of the main drivers for the speciation process, is supported by the niche identity test because the analysis supported significant ecological niche divergence between the two species in the present study. The differentiation between *M. albizona* and *M. bulgardaghica* at the species level was initially explained by morphological characteristics (Nilson and Andren 1985; Nilson *et al.* 1990). However, these species were not separated by molecular data (Stümpel and Joger 2009; Stümpel *et al.* 2016). In contrast to the phylogenetic results, ENM outcomes showed that the ecological requirements of *M. albizona* and *M. bulgardaghica* are different.

The results of the present analysis show that the bioclimatic factors (Bio 11 for *M. albizona* and Bio 7 for *M. bulgardaghica*) that most affect the potential distribution of both taxa are different from each other, and this may significantly affect the differentiation of their niches between the two taxa. Although the differences in the biotic factors of these two taxa, such as life history characteristics and diets, have not been studied yet, these vipers have a remarkable niche difference in abiotic factors in regions close to each other. The results of a recent distributional study revealed that these two taxa could be intertwined in a common region, i.e., there may be a contact zone (Mebert *et al.* 2020). However, in the same study, it was stated that the altitudinal position of this area is lower than where

mountain vipers can exist. This situation might also decrease the gene flow between these two taxa (Mebert *et al.* 2020).

On the other hand, it has been reported that there may be a parapatric contact zone between them, as both taxa have a limited distribution in the Taurus mountains (Schätti *et al.* 1991; Stümpel *et al.* 2016). However, this study's results show no calculated niche overlap between the two taxa in terms of bioclimatic factors and habitat characteristics. In other words, neither this considered contact zone nor any other area met the expectations among these two taxa in terms of identical niches.

According to the EUNIS Habitat Preference analysis, mountain vipers inhabit several types of habitats. Briefly, while *M. albizona* uses 7 unique and a total of 13 EUNIS habitat types, *M. bulghardaghica* does not have any unique habitat solely occupied by itself (supplementary figure 2). Compared with *M. albizona*, *M. bulghardaghica* prefers rockier and/or grassland vegetation. These habitat differences may also have contributed to the ecological niche difference between the two mountain viper taxa.

Since members of the genus *Montivipera* generally prefer mountainous areas as their habitat, they can use very few selected habitats in these fragile and vulnerable ecosystems. Therefore, they are very likely to be affected by environmental changes (Latifi 2000; Rajabizadeh *et al.* 2011; Ansari 2020). The results of the present study proved this. The results showed that *M. albizona* and *M. bulgardaghica* taxa use 13 different habitat types as habitats on the Anatolian Peninsula.

While only 6 of them are used by the taxon of *M. bulgardaghica*, all 13 habitat types can be used by the taxon *M. albizona*. This can be explained by the fact that the *M. bulgardaghica* taxon is distributed in a very narrow area over the eastern Taurus mountains. On the other hand, the land use and geographic distribution of the *M. albizona* taxon are wider than those of *M. bulgardaghica*. Since using different habitat types provides different ecological niches, it may also cause the niche difference between the two taxa in this study. Besides, these results may indicate that *M. albizona* has more tolerance than *M. bulgardaghica* in terms of ecological conditions. *Montivipera albizona* is distributed in wider and more diverse habitats as a result.

Three explanations were suggested for significant niche divergence among species: niche similarity, identity test, and geography (Wellenreuther *et al.* 2012). The results showed a significant ecological differentiation between *M. albizona* and *M. bulgardaghica*, and hence their niches are different. Furthermore, identity test results showed that the niche of *M. albizona* and *M. bulgardaghica* differ significantly. They have been accepted as parapatric species, as it is thought there could be a contact zone between them. (Schätti *et al.* 1991; Stümpel and Joger 2009; Stümpel *et al.* 2016; Mebert *et al.* 2020). However, since the region considered as the contact zone of these two taxa does not have the altitude where they can live, it may have caused them to differ from each other in their evolutionary processes and become different taxa. The results of this study demonstrated that, contrary to the latest phylogenetic results, both taxa are suggested to have different ecological niche requirements.

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Declarations

Conflict of interest The author has no conflict of interest.

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